Tensors

In linear algebra the tensor product of vector spaces V, W is a vector space $V \otimes W$ consisting of elements $v \otimes w$ with identifications:

1) $\lambda(v \otimes w) = (\lambda v) \otimes w = v \otimes (\lambda w)$ for $\lambda \in \mathbb{R}$
2) $(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w$, $v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2$

note that $\dim(V \otimes W) = \dim V \cdot \dim W$

example: $V^* \otimes V^* = \{\text{bilinear forms } V \times V \to \mathbb{R}\}$

Given vector bundles E, F over M, we can take the tensor products of fibers to define

$$E \otimes F := \bigcup_{p \in M} (E_p \otimes F_p)$$

Define: A tensor on M is a section of the bundle

$$T^k_l M := \underbrace{T^*M \otimes \cdots T^*M}_{k} \otimes \underbrace{TM \otimes \cdots \otimes TM}_{l}$$

example: g is a symmetric $(2,0)$ tensor

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We can convert types using

$$g : T^*M \to TM$$

$$g^{-1} : TM \to T^*M$$

With respect to the coord. basis, this works by raising/lowering indices with g_{ij} or g^{ij}

$$\underbrace{a^i \partial_i}_{\text{tangent}} \longmapsto \underbrace{g_{ij} a^i dx^j}_{\text{cotangent}}$$

Volume

From calc., changing variables $x^1 \dots x^n$ to $u^1 \dots u^n$

$$\Rightarrow \quad du^1 \dots du^n = |\det J| \, dx^1 \dots dx^n \qquad J = \left[\frac{\partial u^i}{\partial x^j}\right]$$

Let $g_{ij} = g(\partial_{x^i}, \partial_{x^j})$, $\tilde{g}_{ij} = g(\partial_{u^i}, \partial_{u^j})$

As a tensor,

$$g = g_{ij}\, dx^i \otimes dx^j = \tilde{g}_{ij}\, du^i \otimes du^j$$

$$du^i = \frac{\partial u^i}{\partial x^j} dx^j \quad \Longrightarrow \quad g_{ij} = \tilde{g}_{kl} \frac{\partial u^k}{\partial x^i} \frac{\partial u^l}{\partial x^j}$$

$$\det[g] = \det[\tilde{g}] \, (\det J)^2$$

$\Rightarrow$ The volume density $dV = \sqrt{\det g}\, dx^1 \dots dx^n$ is independent of coord's

Warning: g denotes a tensor, invariant of coord.'s

$\det g$ means $\det[g_{ij}]$ in the coords $x^1, \dots, x^n$

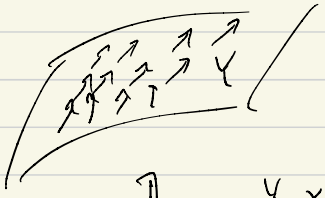
(To make this more confusing, dV is often written as dg!)

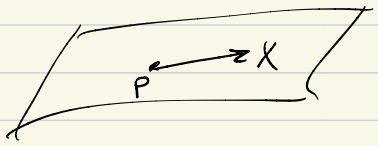
Connections

Consider a vector field Y on a surface $S \subset \mathbb{R}^3$

$Y : S \to TS$ but we can think of $T_pS \subset \mathbb{R}^3$ also

Given $X \in T_pS$ we can choose a curve γ with $\gamma(0) = p$, $\dot\gamma(0) = X$

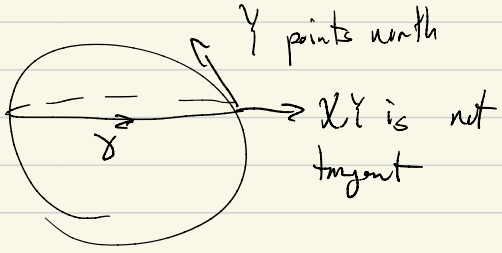




Then $Y \circ \gamma$ is map $I \to \mathbb{R}^3$ which can be differentiated

Then $\frac{d}{dt} Y \circ \gamma \big|_{t=0}$ depends only on X, but is not necessarily tangent to S

$\Rightarrow$ the derivative $X.Y$ does not make sense as a tangent vector



To correct can project:

$\nabla_X Y :=$ orth. projection of $\frac{d}{dt} Y \circ \gamma \big|_{t=0}$ into T_pS

If $\mathcal{T} := \{\text{sections of } TS\}$, then ∇ defines a map $\mathcal{T} \times \mathcal{T} \to \mathcal{T}$

This idea can be generalized to differentiating sections of a vector bundle

Let E be a vector bundle over a mfld M,

$\mathcal{E} := \{\text{smooth sections of } E\}$

$\mathcal{T} := \{\text{vector fields on } M\}$

defin: A <u>connection</u> on $\mathcal{E}$ is a map

$$\nabla : \mathcal{T} \times \mathcal{E} \longrightarrow \mathcal{E}$$

written $X, \phi \longmapsto \nabla_X \phi$ ("derivative of σ in direction X")

satisfying: 1) $\nabla_{(\cdot)} \phi$ is linear over $C^\infty(M)$

$$\begin{cases} \nabla_{X_1 + X_2} \phi = \nabla_{X_1} \phi + \nabla_{X_2} \phi \\ \nabla_{fX} \phi = f \nabla_X \phi \qquad f \in C^\infty(M) \end{cases}$$

2) $\nabla_X(\cdot)$ is linear over $\mathbb{R}$

3) $\nabla_X(\cdot)$ is a derivation, meaning for $f \in C^\infty(M)$

$$\nabla_X(f\phi) = (Xf)\phi + f \nabla_X \phi$$

Check: our definition for $S \subset \mathbb{R}^3$ by projection qualifies as a connection on TS.

Lie bracket of vector fields

Given $X, Y \in \mathcal{T}$ and $f \in C^\infty(M)$

$Yf \in C^\infty(M)$ and so $X(Yf)$ is defined

but XY takes two derivatives $\Rightarrow$ not a vector field

However in $[X,Y] := XY - YX$ the 2nd derivatives vanish

example: in coord's, $X = a^j \partial_j$, $Y = b^k \partial_k$

$$X(Yf) = a^j \partial_j (b^k \partial_k f)$$

$$Y(Xf) = b^k \partial_k (a^j \partial_j f)$$

difference: $[X,Y] = \left[a^j (\partial_j b^k) - b^j (\partial_j a^k) \right] \partial_k$

Thm On a Riemannian mfld M, there is a unique connection ∇ on TM such that

1) $\nabla_X Y - \nabla_Y X = [X, Y]$ (symmetry)

2) $X g(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$ (compatibility)

(This is called the "Riemannian connection")